DAQQA System Features Specification

■■■■■■ — Autonomous Space Health & Garmin Telemetry Ecosystem

Project Track: 2026 NASA Space Apps Challenge • Software for Healthcare Track
Target Platforms: Garmin tactix 7 / fenix 7 (Connect IQ System 7 / Monkey C 4.2+)
Data Standardization: NASA Life Sciences Data Archive (LSDA) & Lifetime Surveillance of Astronaut Health (LSAH)
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EXECUTIVE OVERVIEW: DAQQA (■■■■■■ - Arabic for 'precision') is an end-to-end astronaut health monitoring and predictive clinical decision support ecosystem built for long-duration deep space exploration missions (Lunar Gateway, Mars transit). By deploying on Garmin spaceflight hardware and coupling continuous physiological biometrics (PPG, Pulse Ox) with subjective behavioral check-ins, DAQQA ensures uninterrupted medical vigilance during RF communication blackouts and enables multi-archive data dispatch upon line-of-sight restoration.

1. ■ Cinematic Galaxy & Authentic NASA Planetary Atmosphere

The web interface utilizes authentic photographic textures captured by NASA exploratory missions, creating an immersive aerospace glassmorphism interface with ultra-smooth parallax physics:

Sector / Tab	Celestial Body	Official NASA Mission Source	Active Orbital Beacon Overlay
Tab 1: Mission Simulator	Planet Earth	NASA Blue Marble 2012 (Suomi NPP Satellite)	ISS • 28.5°N (Alpha Station LEO)
Tab 2: Health Intelligence	The Moon (Luna)	NASA Lunar Reconnaissance Orbiter (LRO)	GATEWAY • NRHO (Artemis Relay)
Tab 3: Flight Surgeon HUD	Planet Mars	NASA Viking Orbiter Global Color Mosaic	ARES-1 • TRANSIT (Perseverance Relay)
Tab 4: Dual-Export Hub	Gas Giant Jupiter	NASA Cassini Spacecraft Jovian Flyby	EUROPA • PASS (Juno Clipper)
Tab 5: Monkey C Code	Ringed Giant Saturn	NASA Cassini Equinox True-Color Portrait	CASSINI • GAP (Ring Plane Vector)
Tab 6: Challenge Brief	Cosmic Deep Field	NASA James Webb Space Telescope (JWST NIRCam)	JWST • DEEP FIELD (L2 Halo)

- **Low-Sensitivity Parallax Dynamics:** Utilizes a 60 FPS requestAnimationFrame exponential moving average (LERP) filter with 0.022 sensitivity and an absolute 48px cap to eliminate jitter and visual fatigue during deep clinical review.
- **Celestial Orrery Drawer:** Slide-out cosmic drawer enabling flight surgeons to preview, cycle, and lock planetary views across the mission trajectory.
- **Dark Aerospace Glass Styling:** Custom CSS backdrop filters (.galaxy-glass-card) with cyan vector glows, slate structural borders, and smooth cross-fade opacity transitions.

2. ■ Garmin Smartwatch Hardware Simulation (tactix 7 / f■nix 7)

Interactive tactical hardware housing simulating a Garmin tactix 7 / fenix 7 Spaceflight Edition running Connect IQ System 7:

- **Tactile 5-Button Enclosure:** Functional hardware pushers (LIGHT, UP, DOWN, START/STOP, BACK) with knurled metal grips, carbon lugs, and spaceflight silicone straps.

- **WatchFace HUD Screen:** Animated cardiac pulse indicator, real-time Heart Rate (BPM), SpO2 (%), HRV (ms), analog seconds tick ring, battery monitor, and telemetry link state.
- **In-Flight Behavioral Survey UI:** Astronauts log Sleep Quality (1-5), Sleep Latency (min), Night Awakenings, and POMS ratings (Muscle Fatigue, Mental Focus, Anxiety) directly from their wrist.
- **Rolling Dual-Waveform Sensor Graph:** Live graphical display plotting synchronous PPG (HR) and Pulse Ox (SpO2) rolling waveforms directly on the watch face.
- **Offline Flash Storage Buffer:** Real-time visualization of NVRAM storage capacity, record counts, and burst upload status during RF communication loss.
- **Autonomous Wrist Alert Modal:** Independent on-wrist alarm that triggers haptic feedback and visual emergency warnings during acute hypoxia or severe tachycardia even without Earth communication.
- **Crew Wrist Selector:** Instant switching between all 4 crew members to inspect individual wearable telemetry and local buffer states.

3. ■ Advanced Clinical Health Intelligence & Exertion Analytics

Five specialized mathematical engines delivering personalized, multi-channel physiological surveillance:

A. Synchronous SpO2 & HR Co-Registration (Δt = 0 ms)

Both photoplethysmogram (PPG) Heart Rate and Pulse Oximetry (SpO2) are captured at the exact same millisecond timestamp, enabling authentic cardiorespiratory coupling analysis without temporal lag or phase distortion.

B. HR vs. Physical Activity Concordance (METs & Karvonen Dynamics)

- **NASA Activity Classification:** Categorizes activity levels into Rest/Sleep (1.0-1.5 METs), Light IVA (1.6-2.9 METs), Moderate Work (3.0-5.9 METs), and Vigorous EVA/Exercise (6.0-9.5 METs).
- **Karvonen Expected Heart Rate:** HR_expected = RHR + (HR_max - RHR) * ExertionFraction, dynamically computed from each astronaut's personal resting HR baseline.
- **Exertion Strain Ratio (ESR):** Compares actual HR elevation to metabolic demand to detect Inappropriate Tachycardia at Rest, Exertional Strain Drift, Chronotropic Incompetence, or Superior Aerobic Efficiency.
- **Cardio-Oxygen Saturation Index (COSI):** (SpO2 / HR) * 100, tracking oxygen transport efficiency under operational workloads (normal resting range: 1.40 - 1.70).
- **Nadir 10th-Percentile Resting HR:** Robust moving-window resting baseline calculation filtered strictly during low-activity intervals (METs <= 1.8).

C. Personalized Baseline Anomaly Detection (z-Scores)

Evaluates deviations against an astronaut's learned physiological baselines using adaptive z-scores: $z = (x - \mu) / \sigma$. Eliminates false alarms from generic population averages by alerting only when vitals breach individual standard deviation limits.

D. Sensor Reliability Monitor & Motion Artifact Rejection

Monitors PPG signal quality index (SQI 0-100%), signal-to-noise ratio (SNR in dB), perfusion index, and contact impedance. Fuses 3-axis accelerometer data to automatically discount false-positive spikes during vigorous EVA or exercise.

E. Multimodal Risk Assessment Fusion Matrix (MMRS)

Fuses objective biometrics with subjective behavioral surveys across 5 strain modalities: Autonomic, Cardiovascular, Hypoxic, Behavioral Sleep, and Cognitive. Detects synchronous multi-modality convergence to generate the Composite Multimodal Risk Score (MMRS: 0-100) and automated clinical syndromic diagnoses.

F. Predictive Trend Forecasting (Confidence Cone AI)

Calculates 15-to-60 minute forward time-series linear regressions and velocity rates of change. Renders a 95% confidence cone trajectory providing flight surgeons with early warning before critical decompensation.

4. ■ Ground Station & Flight Surgeon Telemetry Suite

Dedicated mission control monitoring station providing comprehensive crew oversight:

- **Real-Time Lead II ECG Canvas:** 60 FPS HTML5 Canvas rendering Lead II electrocardiogram with authentic P-Q-R-S-T morphology, isoelectric baseline drift, QRS complex, and systolic sweep line.
- **Multi-Channel Continuous Telemetry Charts:** Interactive SVG telemetry charts for Heart Rate, SpO2, HRV, and Stress with NASA safety limit lines and live cursor inspection.
- **Active Crew Biometric Roster:** Complete 4-astronaut crew switcher (Commander Dr. Elena Rostova, Pilot Marcus Vance, Specialist Dr. Sarah Chen, EVA Specialist Alex Rivera) displaying flight day, roles, and status badges.
- **Real-Time Anomaly Triage Detector:** Displays triage cards with severity tags (CRITICAL, WARNING, NOMINAL) and actionable countermeasures.
- **Environmental Life Support HUD:** Continuous tracking of spacesuit pressure (psi), cabin oxygen partial pressure (%), skin temperature (°C), respiration rate, and radiation dosage.

5. ■ NASA Flight Scenario Simulator

Orbital simulation controller allowing flight controllers to trigger 5 mission conditions:

Scenario Protocol	Operational State	Physiological Dynamics	System Response
NOMINAL	Routine Intravehicular (IVA)	Stable baselines, low METs (1.1-1.8), normal COSI	Standard 1Hz telemetry streaming
EVA WORKOUT	Spacewalk / Countermeasure	Elevated METs (7.4), high HR, increased cardiac strain	Motion artifact filtering, concordance tracking
BLACKOUT (LOS)	Orbital Loss of Signal	Nominal to fluctuating astronaut vitals	Autonomous on-wrist NVRAM caching, burst sync upon AOS
ACUTE HYPOXIA	Suit / Cabin O2 Drop	Rapid SpO2 drop (<90%), compensatory tachycardia	Critical wrist haptic alarm, syndromic triage alert
CIRCADIAN DESYNC	Sleep Loss & Shift Work	Low HRV, elevated stress, blunted autonomic recovery	Sleep-behavioral strain weighting, rest prescription

6. ■ NASA Dual-Export Hub & Data Extraction

Bifurcated data distribution architecture complying with NASA standards upon signal acquisition (AOS):

- **Full Telemetry CSV Export:** 34-column CSV matrix compliant with RFC 4180 and UTF-8 BOM, capturing every 1Hz tick, synchronous SpO2, HR, METs, z-scores, SQI, MMRS, and life support readings with zero NaN values.
- **HR vs. Activity Dedicated CSV:** 15-column dataset tailored for cardiorespiratory exercise physiology research.
- **NASA Life Sciences Data Archive (LSDA) Export:** Generates de-identified, pseudonymized research JSON and CSV tables compliant with HIPAA Safe Harbor and NASA IRB Tier 1 Non-PII standards for open academic study.
- **NASA Lifetime Surveillance of Astronaut Health (LSAH) Export:** Encrypted clinical EHR payload with simulated AES-256 GCM credentials for NASA flight surgeons.
- **Official NASA Specification PDF:** Direct download of the 11-page scientific engineering specification (DAQQA_NASA_Telemetry_Specification_and_Thresholds.pdf).
- **Live Patient Clinical Dossier (PDF Print):** One-click printable clinical summary featuring baseline deltas, risk breakdown, and forward trend trajectory.

7. ■ Garmin Connect IQ (Monkey C) Source Code Suite

Production-grade Monkey C architecture developed for Garmin tactix 7 / fenix 7 in microgravity:

- **manifest.xml:** Permissions (Sensor, Storage, Communications, Background) and device targets (tactix 7, fenix 7 Pro, descent Mk3, Forerunner 965).
- **DaqqaApp.mc:** Application lifecycle and background telemetry scheduler.
- **BiometricSensorManager.mc:** High-frequency listener capturing synchronous HR and Pulse Ox streams.
- **OfflineStorageEngine.mc:** Non-volatile RAM storage during communication loss.
- **DualSyncService.mc:** Burst telemetry dispatcher upon re-establishing ground communication.
- **BehavioralSurveyView.mc:** On-wrist survey rendering and tactile button handlers.
- **In-Browser Code Explorer:** Interactive viewer with syntax formatting, copy to clipboard, individual file downloads, and bulk .txt bundle download.

8. ■■ Robustness & Zero-NaN Mathematical Certification

Every mathematical calculation, division, and data export across 21 critical files is defensively guarded with Number.isFinite checks and baseline fallbacks. Verified via an automated Node.js test suite across 100 continuous simulation cycles and string inspection of exported CSV and LSDA tables with zero errors.

Document Verification: Certified Production Architecture	**Verification Status:** Passed (100% Zero-NaN)
NASA Challenge Track: Software for Healthcare Track	**Repository:** github.com/ahmedellakany2006-dotcom/Daqqa